

Module 1: Quantum Fundamentals & Operations

Quantum Computing Workshop

March 20, 2025

What is a Qubit?

- **Classical bit:** 0 or 1
- **Quantum bit (qubit):** Can exist in a superposition of 0 and 1
- Physically implemented using:
 - Electron spin (up/down)
 - Photon polarization (horizontal/vertical)
 - Superconducting circuits (ground/excited states)

Mathematical Representation of Qubits

- A qubit is represented as a vector in a 2-dimensional complex vector space
- Standard basis states:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (1)$$

- General qubit state:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle \quad (2)$$

where α, β are complex numbers $|\alpha|^2 + |\beta|^2 = 1$ (normalization) (3)

Dirac Notation

- $|\psi\rangle$: “ket” - represents a quantum state
- $\langle\psi|$: “bra” - represents the dual (conjugate transpose) of a state
- $\langle\phi|\psi\rangle$: “braket” - inner product between states
- $|\phi\rangle\langle\psi|$: outer product - represents an operator

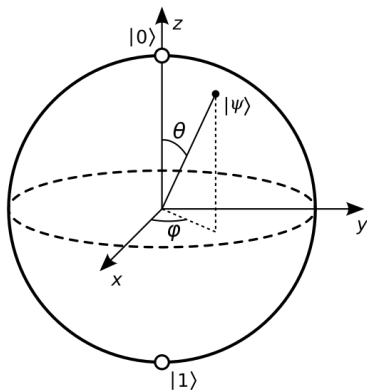
- Measuring a qubit in state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$:
 - Probability of outcome 0: $|\alpha|^2$
 - Probability of outcome 1: $|\beta|^2$
- After measurement, the qubit collapses to the measured state
- Measurement disrupts superposition

The Bloch Sphere

- Geometric representation of a qubit state
- Any pure state can be written as:

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle \quad (4)$$

- θ : polar angle (0 to π)
- ϕ : azimuthal angle (0 to 2π)
- Points on the surface represent pure states
- $|0\rangle$ at north pole, $|1\rangle$ at south pole



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Superposition

- Linear combination of basis states
- Examples:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad (5)$$

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} \quad (6)$$

- Allows quantum computers to process multiple possibilities simultaneously
- Not the same as classical probability (due to interference)

Quantum Gates: Single-Qubit

Pauli-X: Bit flip (quantum NOT gate)

$$X |0\rangle = |1\rangle, \quad X |1\rangle = |0\rangle \quad (7)$$

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (8)$$

Pauli-Y: Bit and phase flip

$$Y |0\rangle = i |1\rangle, \quad Y |1\rangle = -i |0\rangle \quad (9)$$

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad (10)$$

Pauli-Z: Phase flip

$$Z |0\rangle = |0\rangle, \quad Z |1\rangle = -|1\rangle \quad (11)$$

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (12)$$

Quantum Gates: Hadamard

Hadamard (H): Creates superposition

$$H|0\rangle = |+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad (13)$$

$$H|1\rangle = |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} \quad (14)$$

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (15)$$

- Transforms between X and Z basis
- Acts as a square root of NOT in some contexts

Multi-Qubit Gates: CNOT

CNOT (Controlled-NOT):

- Two-qubit gate: control and target
- Flips target qubit if control qubit is $|1\rangle$

$$\text{CNOT } |00\rangle = |00\rangle \quad (16)$$

$$\text{CNOT } |01\rangle = |01\rangle \quad (17)$$

$$\text{CNOT } |10\rangle = |11\rangle \quad (18)$$

$$\text{CNOT } |11\rangle = |10\rangle \quad (19)$$

- Essential for creating entanglement

Entanglement

- Quantum correlation stronger than any classical correlation
- **Bell States:** Maximally entangled two-qubit states

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \quad (20)$$

$$|\Phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}} \quad (21)$$

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}} \quad (22)$$

$$|\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}} \quad (23)$$

- Measuring one qubit instantly determines the state of the other
- No classical information transmission (no-communication theorem)

Key Concepts Review

- **Qubits:** Quantum bits that can exist in superposition
- **State Vector:** Mathematical representation ($\alpha |0\rangle + \beta |1\rangle$)
- **Bloch Sphere:** Geometric visualization of qubit states
- **Quantum Gates:** Manipulate qubit states (X, Y, Z, H, CNOT)
- **Measurement:** Collapses superposition with probabilities $|\alpha|^2$ and $|\beta|^2$
- **Entanglement:** Quantum correlation created with multi-qubit gates

Next Steps

- Quantum teleportation protocol
- Understanding quantum algorithms
- Implementing Deutsch's algorithm